بسم الله الرحمن الرحيم

OOGENESIS COMPARISON OF GAMETES

LEARNING OBJECTIVES

At the end of the lecture, student should be able to,

- Define the term oogenesis.
- Describe the process of oogenesis.
- Differentiate between primary and secondary oocytes.
- Compare the male and female gamete.

DEFINITION OF OOGENESIS

- **Oogenesis (ovogenesis)** is the sequence of events by which oogonia are transformed into **mature ovum**.

- **Oocyte** (*L. ovum*, egg)

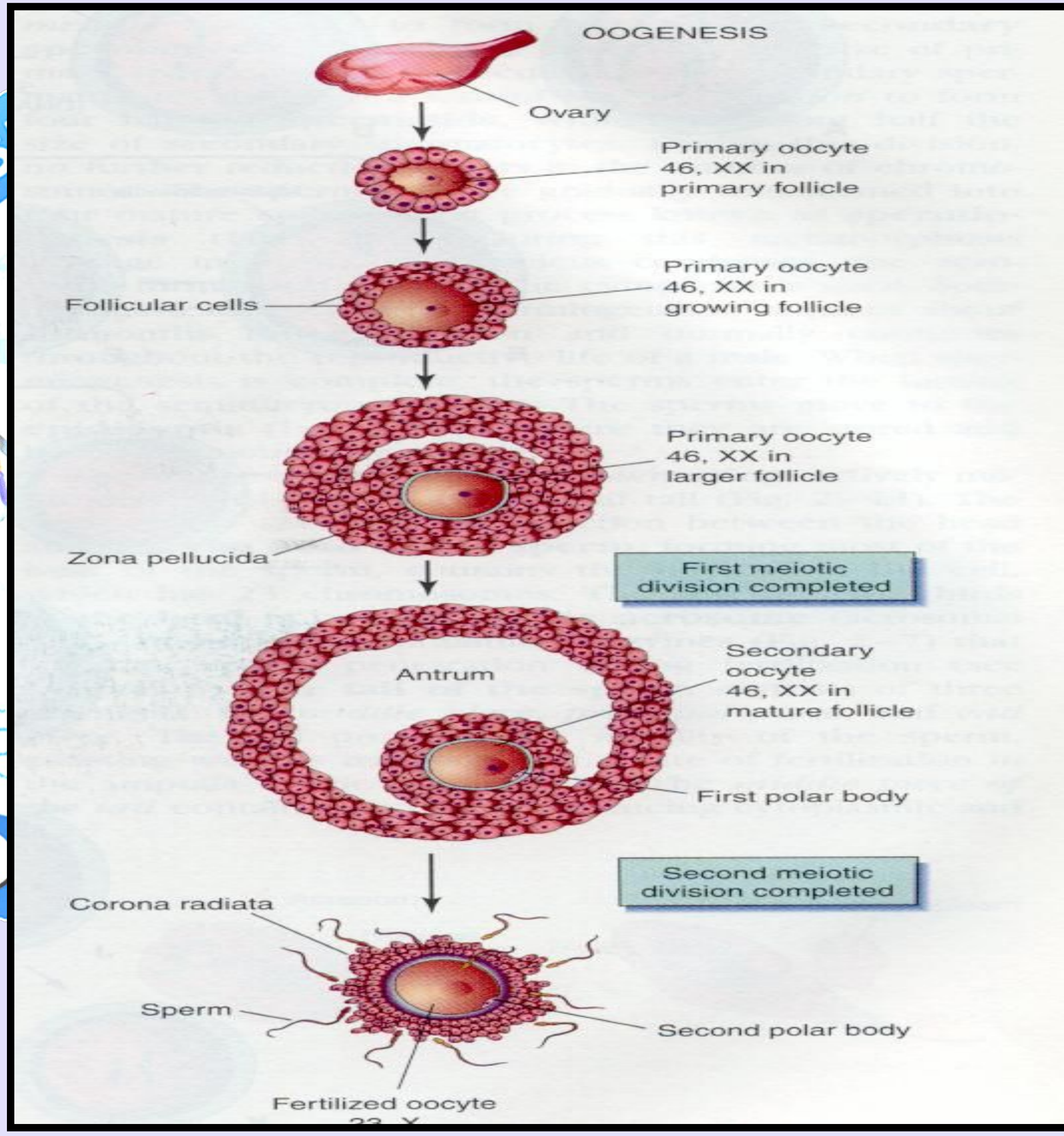
The female germ or sex cells are produced in the *ovaries*. When mature, the oocytes are called secondary oocytes or mature oocytes.

Duration/time period of oogenesis

- The maturation process of Oogenesis begins before birth and is continuous throughout female reproductive life.
- Oogenesis continues to *menopause*, which is permanent cessation of the *menses* (bleeding associated with the menstrual cycles).

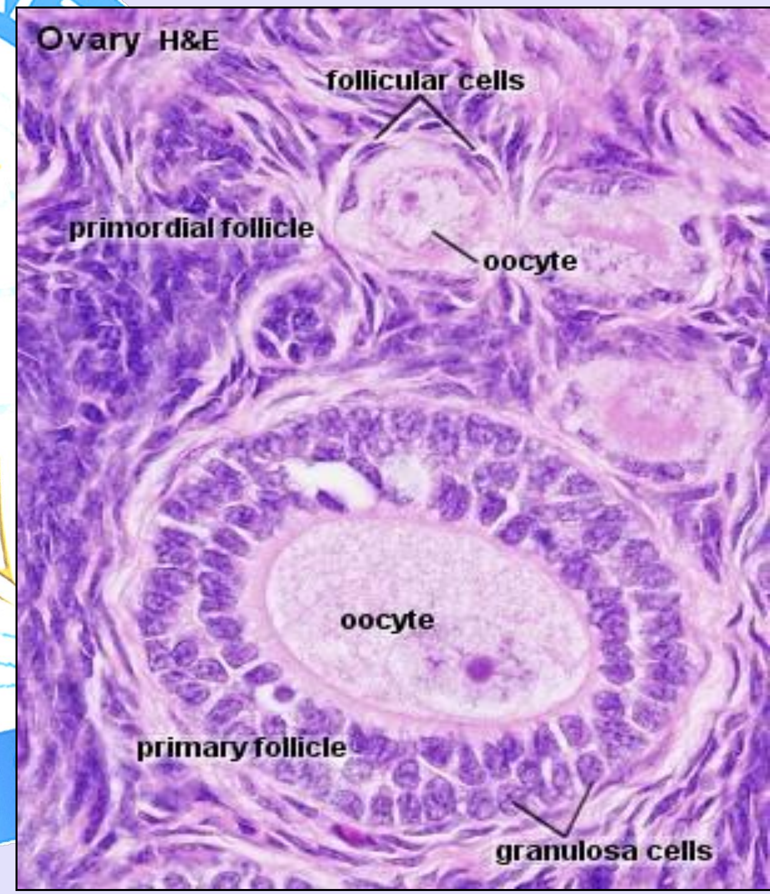
Prenatal Maturation of Oocytes

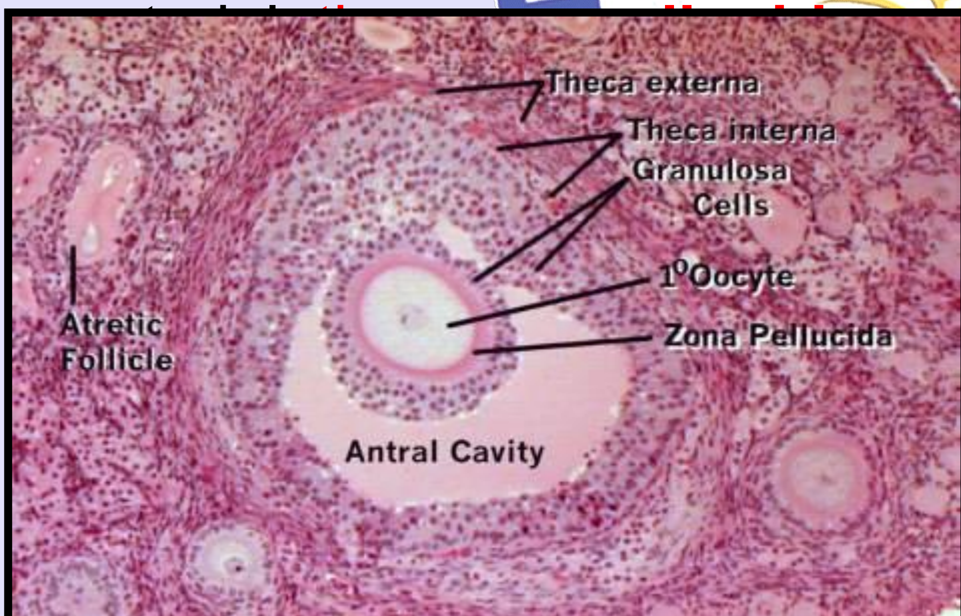
- During early fetal life, **oogonia divided by mitosis**. *Oogonia enlarge to form primary oocytes before birth*
- As a **primary oocyte** forms, connective tissue cells surround it and form a single layer of flattened, follicular epithelial cells
- The primary oocyte enclosed by this layer of cells constitutes a **primordial follicle**



Prenatal Maturation of Oocytes

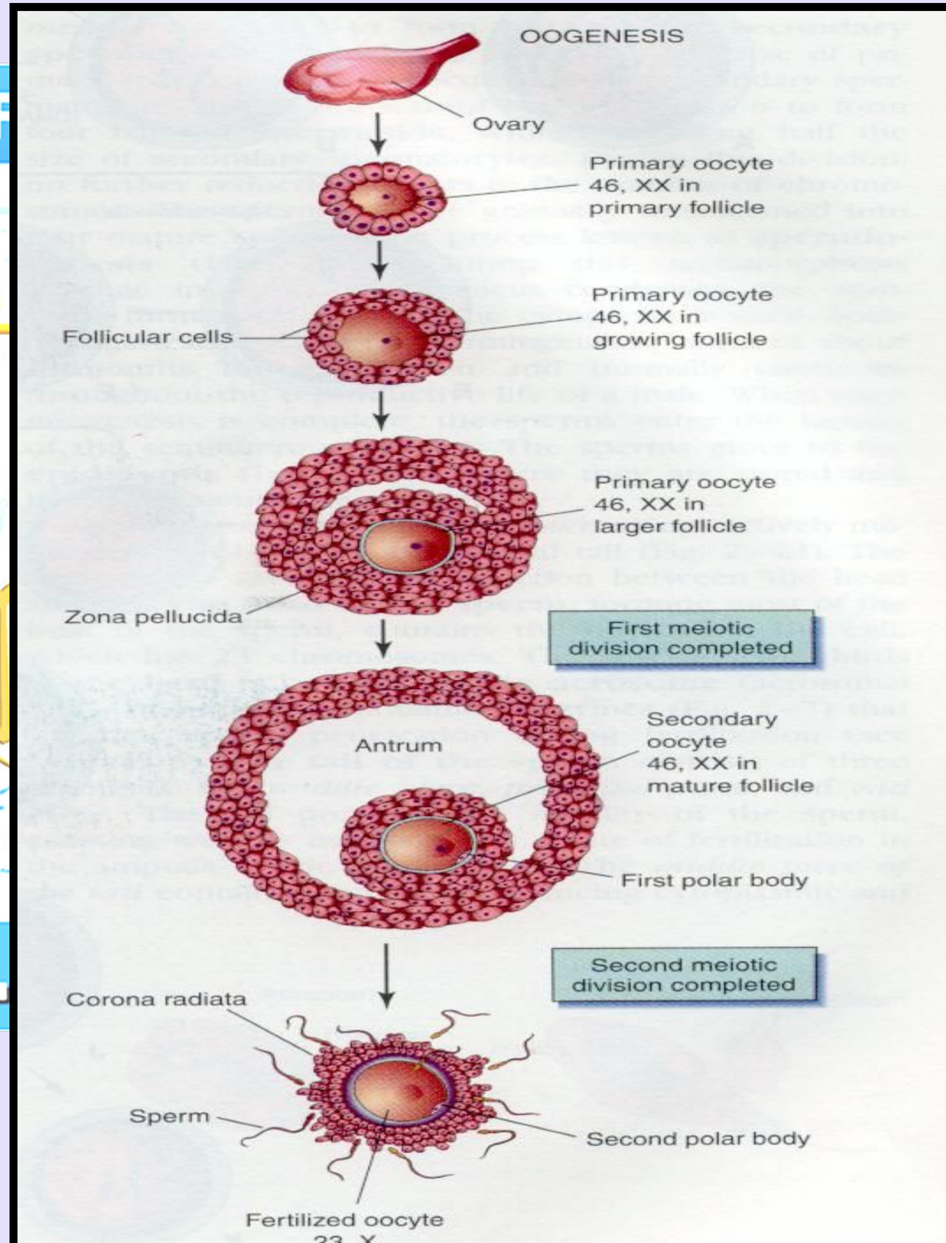
- As the primary oocyte enlarges during puberty, the follicular epithelial cells become cuboidal in shape and then columnar, forming a **primary follicle** .
- The primary oocyte soon becomes surrounded by a covering of amorphous acellular glycoprotein





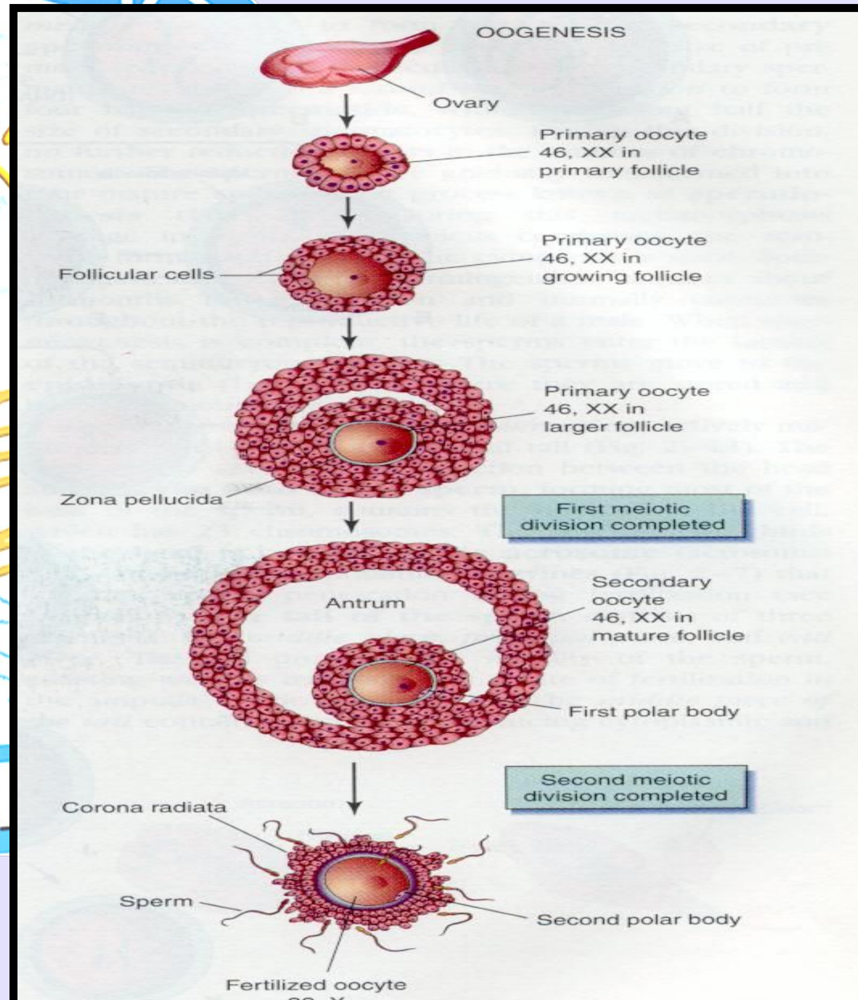
Prenatal Maturation of Oocytes

- **Primary oocytes begin the first meiotic division before birth, but completion of prophase I does not occur until adolescence.**
- The follicular cells surrounding the primary oocyte are believed to secrete a substance, *oocyte maturation inhibitor*, which keeps the meiotic process of the oocyte arrested.
- Beginning during puberty, usually one follicle matures each month and ovulation occurs.



Postnatal Maturation of Oocytes

- The primary oocytes in suspended prophase (dictyotene) are vulnerable to environmental agents such as radiation.
- The primary oocytes remain dormant in the ovarian follicles until puberty. As a follicle matures**, the primary oocyte** increases in size and, shortly before ovulation, completes the first meiotic division to give rise to a **secondary oocyte** and the **first polar body.**

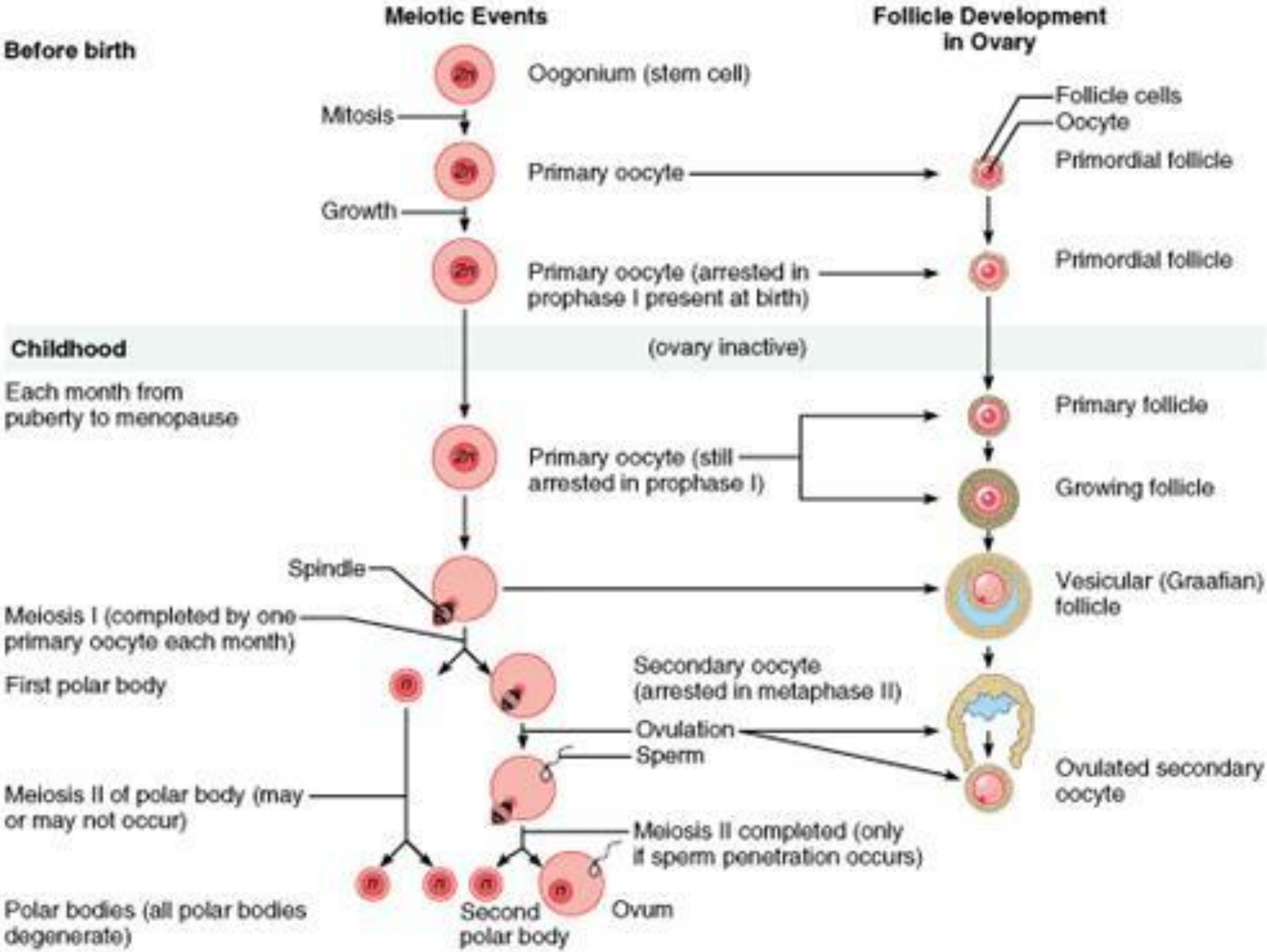


Postnatal Maturation of Oocytes

- The **secondary oocyte** receives almost all the cytoplasm , and the first polar body receives very little. The polar body is a small, nonfunctional cell.
- **At ovulation,** the nucleus of **the secondary oocyte enter in to the second meiotic division, but progresses only to metaphase,** where division is arrested.
- If a sperm penetrates the secondary oocyte, the second meiotic division is completed, and most cytoplasm is again retained by one cell, the **fertilized oocyte**.
- The other cell, the **second polar body**, also a small nonfunctional cell.

Postnatal Maturation of Oocytes

- *There are approximately two million primary oocytes in the ovaries of a newborn female*,
- Most regress during childhood so that by adolescence no more than 40,000 remain.
- Of these, only approximately 400-500 become secondary oocytes and are expelled at ovulation during the reproductive period.
- Few of these oocytes, if any, are fertilized and become mature.



Meiotic Events
Follicle Development in Ovary
Before birth
2n
Oogonium (stem cell)
Mitosis
Follicle cells
Oocyte
Primordial follicle
2n
Primary oocyte
Growth
Primordial follicle
2n
Primary oocyte (arrested in prophase I present at birth)
Childhood
(ovary inactive)
Each month from puberty to menopause
Primary follicle
2n
Primary oocyte (still arrested in prophase I)
Growing follicle
Spindle
Vesicular (Graafian) follicle
Meiosis I (completed by one primary oocyte each month)
First polar body
n
Secondary oocyte (arrested in metaphase II)
Ovulation
Sperm
Ovulated secondary oocyte
Meiosis II of polar body (may or may not occur)
Meiosis II completed (only if sperm penetration occurs)
n
Polar bodies (all polar bodies degenerate)
Second polar body
Ovum

Comparison of male and female gametes

- ***The oocyte is a massive cell*** compared with the sperm and is immotile, whereas the microscopic sperm is highly motile.
- The oocyte is ***surrounded by the zona pellucida*** and a layer of follicular cells, *the* ***corona radiata.***
- The oocyte also has an ***abundance of cytoplasm containing yolk granules (griffian follicle),*** which may provide nutrition to the dividing zygote during the first week of development.
- With respect to sex chromosome constitution, there are ***two kinds of normal sperm:*** 22+ X and 22+ Y, whereas there is only ***one kind of normal secondary oocyte***: 22+ X .
- *The difference in the sex chromosome complement of sperms forms the basis of primary sex determination.*

Comparison of male and female gametes

Similarities

- Arise from epithelial germ cells (Epiblast).
- Start with cell division via mitosis
- Involve cell growth before meiosis
- Involve 2 rounds of cell division (meiosis) to produce haploid cells
- Involve differentiation

Comparison of male and female gametes

Spermatogenesis only

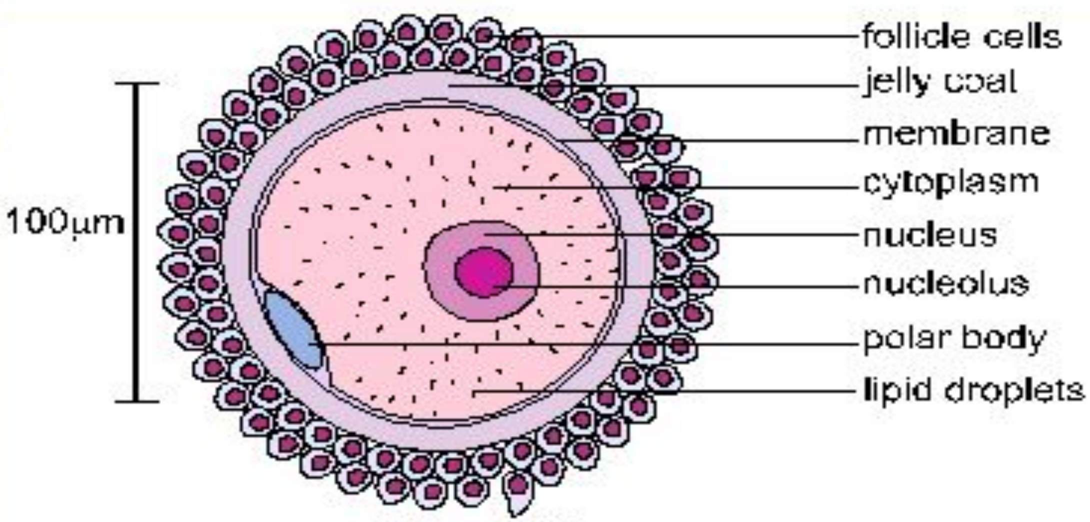
- Nearly unlimited number of sperm can be produced
- Four sperm produced from one spermatogonium.
- Sperm formation begins at puberty
- Sperm form daily
- Sperm release is under voluntary control
- Sperm production may continue up to the old age

Oogenesis only

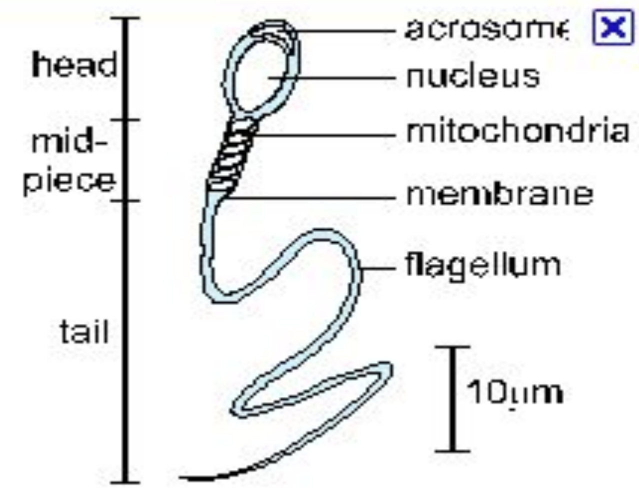
- Limited number of eggs can be produced
- Only 1 egg produced from one oogonium.
- Oogenesis begins in utro.
- Ova mature once a month during puberty
- Ovum release is not under voluntary control; (midpoint of menstrual cycle)
- Ovum production terminates at menopause

Comparison of male and female gametes

	Ovum	Sperm
Size	Over 100µm in diameter.	About 25µm long
Number produced	Relatively few and aren't replenished.	Trillions produced in a lifetime, and 400 million per ejaculate, constantly being developed.
Mobility	The ovum gets moved along the Fallopian tube by cillia.	The sperm move themselves with the tail.



A Human Ovum



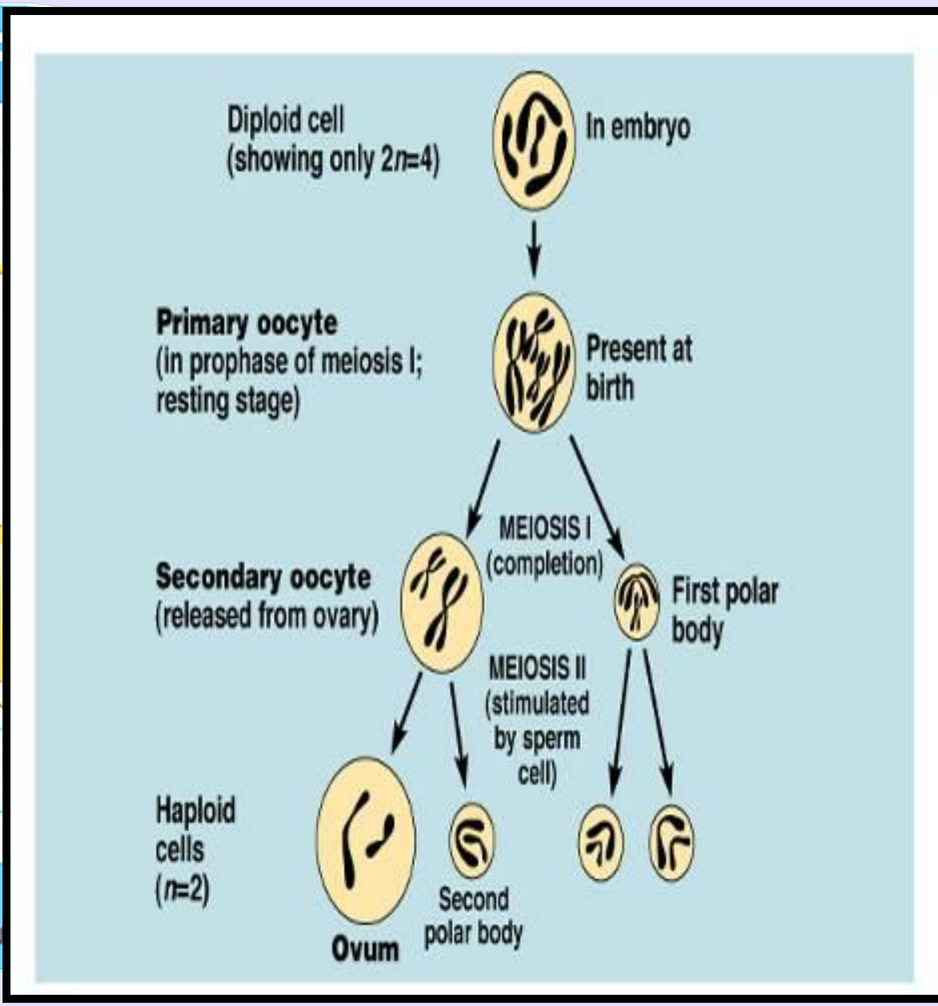
A Human Sperm

Abnormal Gametes

- During gametogenesis, homologous chromosomes sometimes fail to separate. As a result of this error of meiotic cell division-**nondisjunction**-some gametes have 24 chromosomes and others only 22.
- If a gamete with 24 chromosomes unites with a normal one with 23 chromosomes during fertilization, a zygote with 47 chromosomes forms . This condition is called **trisomy** because of the presence of three representatives of a particular chromosome instead of the usual two.
- If a gamete with only 22 chromosomes unites with a normal one, a zygote with 45 chromosomes forms. This condition is known as **monosomy** because only one representative of the particular chromosome pair is present.

Summary; Oogenesis

- Occurs in the ovaries
- Formation of oocytes, eggs
- **Begins prior to birth!**
  - Spans all of pre-reproductive & reproductive life
- **Begins with one cell**
  - Which becomes two; Then finally, one mature ovum & three polar bodies (ultimately degenerate).



Summary; Oogenesis

A. Oogonia

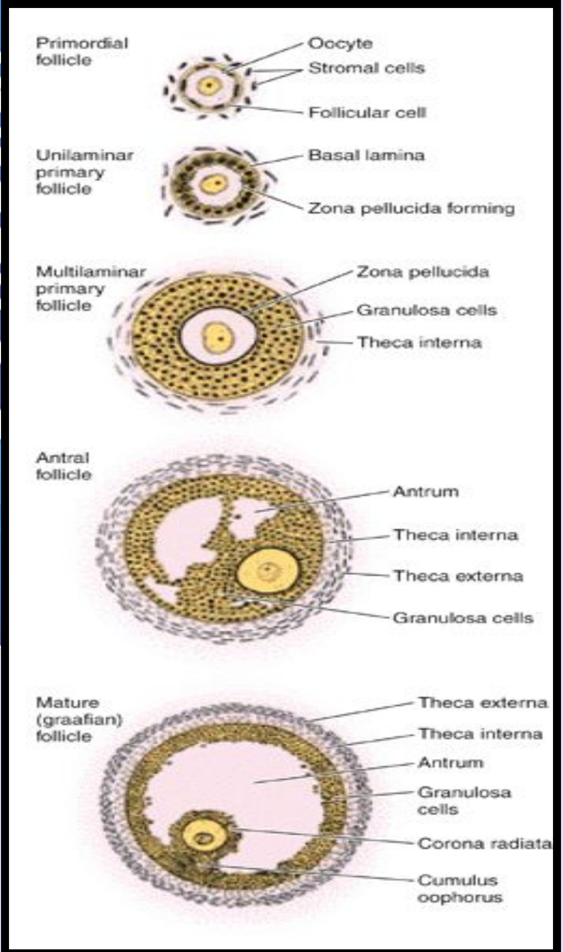
1. Develop from epiblast during 2^{nd} week.
2. migrate to ovary from wall of yolk sac near allantois during the 4^{th} week.
3. mitosis continues up to 5 months.

B. Primary oocytes

1. prophase of 1st meiotic division.
2. 3rd-7th months.

C. Secondary oocyte

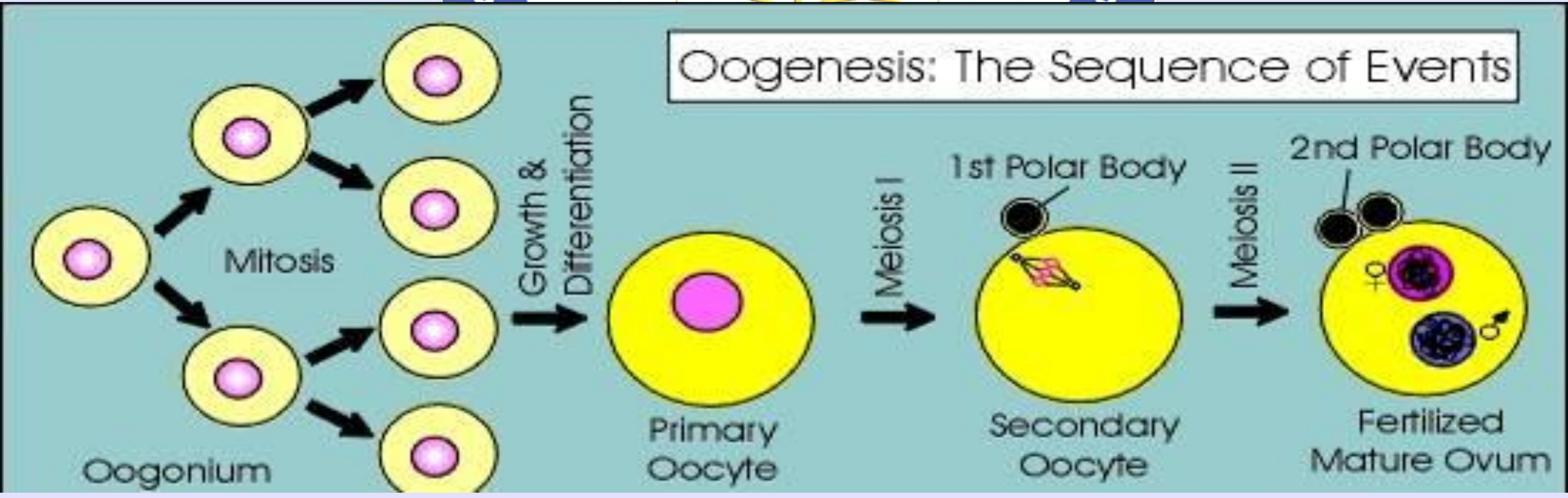
4. 2^{nd} meiotic division began at puberty
5. just before ovulation
6. arrested in metaphase II
7. second polar body + oocyte
8. ovum viable for 24-48 hrs.



Summary; Oogenesis

D. Second meiotic division

1. complete only after fertilization
2. second polar body + ♀ pronucleus
3. zygote = ♂ + ♀ pronuclei fuse
4. mitotic div.




THANK
YOU!